Superresolution CT

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Summary of Technology

Superresolution CT is a technique which focuses on obtaining high-resolution images (x1000) from low resolution images, all while using a small radiation dose. The algorithm is being used to reconstruct blurry images into clear ones, this is done using a UNet neural network, which is a type of basic machine learning algorithm.

Medical Problem Being Solved

The current aim of developing such technology is for research purposes. The goal is to be able to look at smaller structures in more details. An example is looking at the orientation of muscle fibres inside the heart. This orientation is not random, it follows a certain pattern which cannot be seen using traditional CT imaging techniques, (due to limitations in the technology's spatial resolution). This orientation is crucial for the heart as it is what makes the muscle contract correctly. If fibres inside the heart are misaligned or disorganized (due to cardiac disorders or traumatic events), then they will disrupt the electrical and mechanical properties of the organ, preventing it from pumping blood correctly inside the human body.

Conditions such as myocardial fibroses are also related to fibre organization. This condition is a build-up of scar tissue inside the heart, which leads to an impairment of the heart's contracting abilities. It prevents the blood from being pumped correctly and can lead to heart failure. (CDC, 2023)

As of 2023, around 6.7 million people above the age of 20 in the United States are affected by heart failure. This number is expected to increase to 8.5 million by 2030, making it an urgent healthcare concern. From a financial perspective, heart failure is estimated to cost hospitals between $39.2 to $60 billion dollars. This number is estimated to reach $70 billion in 2030. (Bozkurt, 2023)

Currently, there is no direct treatment for myocardial fibrosis which is only one possible cardiac condition which lead to heart failure. Thus, developing technologies such as superresolution CT can contribute in the research towards treatments which can prevent heart failure. (Yetman, 2023) (Heidenreich et al., 2022)

Patient Statistics

In the USA, more than 70 million CT scans are performed every year, each costing from $300 to $3,000. The question to ask now is, why CT scans instead of MRI, which tend to have better resolution and use no ionising radiation? A crucial factor to consider when studying heart conditions is that many patients use medical devices such as pacemakers, which are incompatible with the MRI technology (as they are made of metal). In fact, about 3 million Americans use pacemakers to prevent heart failure. Therefore, developing CT superresolution techniques could allow millions of patients to receive medical scanning without any risks of injury to the patient or damage to the pacemaker. (Bosch de Basea et al., 2015) (Imaging, 2017) (Schoenfield, n.d.)

Uniquity

The technology itself is simple to implement and use. It also is not only applicable to cardiovascular studies, but another example is also glomeruli distribution (small network of blood vessels responsible for filtering fluids in the body). As it has a much higher resolution than anything else currently available on the market, it revealed the small structures that were previously unspecified.

Market

1. Market for Imaging

As the aging population grows, as does the market for medical technologies to aid clinicians in diagnosis and treatment, specifically in terms of medical imaging. At $16.3 billion in 2022, the medical imaging industry is predicted to develop at a compound annual growth rate (CAGR) of 8.2% by 2030. As of 2022, computed tomography (CT) scans accounted for the greatest portion of this industry, at 25.2%. Many researchers are concentrating on high-resolution imaging that does not expose patients to large doses of ionising radiation. (GVR, 2021)

Creating high-definition images (about 3840 X 2160 pixels) with 4K medical imaging specifically offers the advantage of increasing detail and producing minimally invasive imaging. With a CAGR of 12.3%, this market is predicted to reach $2.8 billion by 2032. Leading companies in the research include Sony and Olympus. (PMI, 2023)

2. Market for cardiovascular disease treatment

As cardiovascular disease is the leading cause of death each year, there is a large market for treatment related to this condition. With a CAGR of 3.76%, investment in cardiovascular research and clinical work reached $0.44 trillion in 2022, with millions of people being diagnosed and dying from heart disease. One example is myocardial fibrosis, which was marketed for $765.4 million in 2022, and is expected to rise at CAGR of 6.1% between 2023 and 2030, reaching $1.6 billion. (Statista, 2023)

3. Buyers

Hospitals and Medical Centres
Target: Radiology Departments, Imaging Centres.
Why: Hospitals and medical centres are primary users of advanced medical imaging equipment. Radiology departments use CT scanners extensively for diagnostics and patient care.

Private Practices and Specialized Clinics
Target: Private clinics, specialty practices
Why: Smaller healthcare facilities may also invest in advanced imaging technology to enhance their diagnostic capabilities. Cardiology clinics have a particular interest in specific cardiovascular diseases such as myocardial fibrosis and will invest in simplifying the diagnosing process and treatment process.

Diagnostic Imaging Service Providers:
Target: Companies providing diagnostic imaging services
Why: Companies specializing in diagnostic imaging services may seek state-of-the-art equipment to offer advanced imaging solutions to various healthcare facilities.

4. Market Strategies

This technology will be demonstrated to physicians, mostly in the US, at medical conferences. As it will help them to better treat cardiovascular disease as well as lessen their workload, saving money and time.

Intellectual Property

UNet and most of the algorithm use information that is available on the public domain. Furthermore, having researched patent options, it was found after using the term "*super resolution*" that a patent already existed under Adobe Inc. Another search was conducted using the words "*superresolution CT*" and a patent under the Taiyuan University of Science and Technology in China existed already too. Although patent rules may differ slightly between China and the USA, there is a lot of uncertainty on how much intellectual property we have the right to. (Zhang, Lin, and Wang, 2018) (Xiong Zhang et al., 2019). A potential option would be to patent the snippets of the algorithm which are original, however this could cost around $8,000 to $10,000 (Urzedowski, 2022) all while still have uncertainties of ownership (and being sued). A final solution is that our specific algorithm will be copyrighted and therefore cannot be directly copied. On top of this, the algorithm would not be released to the public domain, and would not be public knowledge, with a cybersecurity employee hired to ensure secrecy. In terms of the Chan Zuckerberg Initiative, it is a charity and does not have any ownership on this technology.

Achievements to date

This medical technology is in the preliminary stages of development, specifically stage 2 of the Technology Readiness Level. The concept has been devised and details of it have been planned out, distinct groups have been organised, from people working on the algorithm, to scientists taking images of organs to be used for testing. With their funding organised by a charity group from Facebook's Mark

Zukerberg. The next stage is to ensure that the high-resolution images generated have the correct general information, such as diseases and structures.

Regulatory path

The device is a software, which is meant to be used by the manufacturer to be used for one or more of the medical purposes, therefore falls under the category of medical device. Under the EU laws (19).

Item	**Comparator device:** UMR 680 (magnetic resonance diagnostic device) Uses super-resolution	**New device:** Super-resolution CT	Types of change and impact to safety and effectiveness
Detection target	Head and body (including heart)	Head and body (including heart)	Same
Intended use	Capable of constructing sagittal, transverse, and cross-sectional images, and that display internal anatomical structure	Assisting radiologist in obtaining a high-resolution image from an inadequate quality of a low-resolution scan	Both are intended to provide a detailed visualization of the image using AI enhancing algorithms
Indications of use	To observe internal anatomical structure Helps in diagnosis. Uses a deep learning-based image processing called Deep Recon for de-noising and create image super-resolution processing.	To scan structures such as heart fibers. The software could aid radiologist to get a better resolution scan from a low dose scan. Able to detect internal anatomical structure in detail	Both uses superresolution to aid with the image creation. Superresolution CT provides a more detailed visualisation of vascular structures in the heart. CT scanning occurs faster and its cheaper

Risks of harm that might arise from this technology

The research team, led by University of Cambridge, found that there were problems including mistakes and artefacts in the final photographs that were not present in the non-AI supported methods (Antun et al 2020). The issues were present across several medical neural networks.

The National academy of sciences suggests that AI-supported imaging for diagnosis, should not be solely used for diagnosis as it showed instability. Because the AI algorithms reconstructed the scan using previous data, they are susceptible to slight changes in the input, which occasionally resulted in a reconstruction with missing detail. In addition, the image that is produced in a real-life scenario might differ, due to real-world medical imaging involving more complex and additional factors. With all the negatives considered, there will not be any harm as the super-resolution is used to assist in diagnosing and there will always be clinician input. (Antun et al 2020).

Probable data required to demonstrate the device's efficacy and safety for its intended use

- Data from the comparator device shows that you would need a sample from 264 volunteers to train neural network with about half and half of each gender
- Performance testing if it were to be commercialised would require around 68 US subjects
- Information from the database from the "*human organ atlas project*" for retrospective clinical trials
- Before commercialization of the device, the FDA would need to approve that the device meets some performance clearance requirements that are set by us. Which is resolution (about 2.6um voxel size), image contrast resolution, noise and uniformity and image artifacts.

Time to market:

- FDA approval would need be required, which would take from one week to 8 months depending on whether the device has been through premarket approval

- Pre-market approval would require scientific evidence to assure that the device is safe and effective for its intended use (2 months). The application is made by the manufacturer
- The algorithm software for researching and aiding the radiologist during scans, (according to the U.S. FDA classification system) would fall under class 2 medical device (medium risk)
- Must go through quality management system afterwards (ISO-13485) to ensure quality in the development and manufacturing stage (5 months)
- Class 2 devices would require submitting a pre-market submission (Premarket notification (510k)). This is where it is demonstrated that a similar predicate device, which works in similar fashion, is cleared by the FDA. It could take up to 6 months to get regulatory clearance.
- Lastly post-market surveillance (PMS) responsibilities are a continuous monitoring of the safety and performance after they have been released
- 1.5 years of retrospective clinical trials + 6 months to file the documentation + 8-month approval would total to about 3 years (Fenton, 2021)
- If there are plans to expand to other market it would have to go through the international regulatory compliance such as the medical device regulation (MDR). Filling of a document showing that it adheres to the MDR is required. Other details such as language translation and different labelling within EU countries need to be considered (Science Business, 2023). This process would take an additional year

Team requirements

Software engineers are responsible for the complete software development lifecycle. They analyse user requirements, design, and plan the structure and functionality of software systems, and then proceed to code and debug the software using programming languages. Their tasks also involve optimizing software performance, ensuring security considerations in code writing, and implementing version control to manage and track changes.

Cybersecurity professionals are pivotal in securing computer systems, networks, and data from unauthorized access and potential threats. Their tasks include patching vulnerabilities, deploying defence mechanisms, conducting malware analysis, managing authentication, and ensuring data encryption.

Legal consultants (Quality & Regulatory Legal and IP) ensure the company's adherence to regulations and quality standards in Quality & Regulatory. Simultaneously, they handle legal affairs and intellectual property issues in Legal and IP, managing contracts, intellectual property assets, and providing legal counsel.

Accounting consultants contribute to Commercial (Sales/Business Development) by providing financial data and analysis to support sales initiatives, participating in pricing and contract negotiations, and offering financial advice for business expansion. Their responsibilities span ensuring financial health, compliance, and providing financial support for business development.

The CEO leads and manages the senior executive team, making strategic decisions across key areas such as finance, operations, and marketing. They ensure effective implementation of decisions and communicate regularly with the investors to report on the company's operational status and financial performance, fostering adherence to best governance practices.

Radiologists specialize in interpreting medical images such as X-rays, CT scans, MRI scans, and ultrasound to diagnose a variety of medical conditions, including tumours, fractures, infections, and organ diseases. They may contribute to treatment planning, especially in radiation therapy, and provide image guidance for surgeries. Additionally, radiologists engage in medical research to enhance diagnostic imaging techniques and treatment methods.

Post-market surveillance encompasses continuous monitoring and analysis of a product's safety and performance after its market launch. This involves addressing adverse events, ensuring regulatory compliance, and implementing measures to enhance product quality.

Quality Regulatory Managers ensure products meet regulations and quality standards. They manage compliance, quality control, and regulatory strategies, ensuring adherence to laws throughout the product lifecycle.

Estimated Team Cost

Estimate of gross yearly salary for high skill-level employees:

- Software engineer: $87.5k
- CEO: $150k
- Manager: $150k
- Cybersecurity: $80k
- Radiologist: $33k
- Post market surveillance: $75k
- Sales:50k
- Legal: $110k
- Accounting: $63k
- Quality regulatory manager: $103k

Total yearly salary costs:

- Year 1-3 salaries: $1,477.5k (Manager, cybersecurity, 3 x software engineers)
- Year 4 salaries: $492.5k (CEO, cybersecurity, 3 x software engineers)
- Year 5-6 salaries: $1,117k (CEO, cybersecurity, 3 x software engineers, 2 x radiologists)
- Year 7-9 salaries: $2,019k (CEO, cybersecurity, 3 x software engineers, 2 x radiologists, post market surveillance, legal advisor, accounting advisor, quality regulatory manager, 2 x sales)

Additional costs:

- Space: $50,000 (annually)
- Expenses: $20,000 (annually)

Total cost: (years 4-9, in which investments will be made) = $4,048,500 ≈ $4.1 million

Analysis for Investibilty

It can be approximated that the technology would need 9 years to be developed and put on the market. The suggestion is to wait 3 years for the software to be completed. If this has potential, an investment of $562k is needed. This leaves year 4 for clinical trials in which it will be seen whether the device works with living organisms or not. If it does, $3.5 million (approximately) is needed between years 5 to 9. After being available on the market for a year, if we assume a profit of $50 per CT scan, (out of $300 CT cost), for ~ 50,000 patients (out of the 70 million who get CT scans), then it would make a revenue of $2.5 million (but no profit yet). After two years on the market, this turns a profit of $0.9 million, and from year 3 onwards, profits would be $2.5 million with 10% annual growth rate as seen from the market growth prediction for CT (GVR, 2018). Therefore, in the most optimistic case where it works on patients and that intellectual property is not a major issue, this technology has potential. If, however, after year 4, the technology is still only functioning for dead organisms, then this obstacle, along with the IP issues, and the fact that the algorithm may cause diagnosis complications due to inaccurate image reconstructions, then the technology will be limited to research purposes only and would be deemed not worth investing more money in.

Conclusion

In conclusion, having considered all factors (IP, healthcare benefits, technical obstacles, finances), the team believes the technology is not worth investing in.

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